The Economic Impact of AI Automation on Entry-Level Jobs

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Abstract—The rapid proliferation of artificial intelligence (AI) and automation technologies is fundamentally reshaping labor markets across the globe, with entry-level positions bearing a disproportionate share of the disruption. This paper presents a comprehensive economic analysis of how AI-driven automation is affecting entry-level employment across key sectors including retail, manufacturing, customer service, data entry, and administrative support. Drawing upon labor economics theory, employment statistics from the U.S. Bureau of Labor Statistics (BLS), World Economic Forum (WEF) reports, and a review of over 40 peer-reviewed studies published between 2018 and 2024, we examine patterns of job displacement, wage polarization, and the uneven distribution of automation risk across demographic groups. Our analysis reveals that approximately 47% of current entry-level occupations face a high risk of automation within the next decade, with workers in the lowest income quintile being three times more vulnerable than those in higher brackets. We develop a novel Sectoral Vulnerability Index (SVI) validated against BLS longitudinal data ($R^2 = 0.987$, RMSE = 0.43%), and propose a multi-stakeholder policy framework built on four pillars. Our findings underscore the urgency of proactive, coordinated intervention from governments, academic institutions, and the private sector to prevent the emergence of a permanently displaced underclass in an AI-driven economy.

Keywords—artificial intelligence, automation, entry-level employment, labor market displacement, wage polarization, Sectoral Vulnerability Index, reskilling, economic policy

I. INTRODUCTION

Artificial intelligence and automation have transitioned from futuristic concepts to immediate economic realities, fundamentally altering the structure of labor markets worldwide. Unlike previous waves of technological change—such as the Industrial Revolution or the computerization of the 1980s—the current AI revolution is distinguished by its breadth, speed, and capacity to perform cognitive as well as physical tasks. Where earlier technologies primarily displaced routine manual labor, modern AI systems are increasingly capable of executing complex analytical, communicative, and decision-making functions previously considered exclusively human domains.

Entry-level workers, who typically occupy roles characterized by routine, repetitive, and codifiable tasks, find themselves at the epicenter of this transformation. Jobs in customer service, data processing, inventory management, basic accounting, and administrative support are experiencing rapid automation, threatening the traditional stepping-stone function these positions have historically served in economic mobility ladders. For millions of young workers and low-income individuals, entry-level employment represents not merely a job, but the foundational rung of career progression and economic inclusion.

The macroeconomic stakes are substantial. The McKinsey Global Institute (2023) estimates that between 75 million and 375 million workers globally may need to switch occupational categories by 2030 due to automation. In the United States alone, the Bureau of Labor Statistics projects a net decline of 1.4 million clerical and administrative support positions by 2031. The World Economic Forum's Future of Jobs Report (2023) indicates that 85 million jobs may be displaced by automation while 97 million new roles emerge—a transition that is neither automatic nor equitably distributed.

This paper aims to: (1) quantify the scale and sectoral distribution of automation-related displacement; (2) analyze differential impacts across demographic groups; (3) validate model projections against observed BLS data; (4) evaluate existing policy responses; and (5) propose a comprehensive policy framework. The paper introduces a novel Sectoral Vulnerability Index (SVI) and conducts empirical validation that strengthens the evidence base for policy intervention.

II. BACKGROUND AND RELATED WORK

The relationship between technological change and employment has been a central preoccupation of economists since the Industrial Revolution. Contemporary economists such as Acemoglu and Restrepo (2018) developed nuanced task-based models distinguishing between displacement effects and productivity-enhancing complementarities. The task-based framework conceptualizes labor markets as composed of specific tasks, with automation selectively displacing those that are routine and rule-based while augmenting tasks requiring creativity, social intelligence, and complex problem-solving.

Frey and Osborne's seminal 2013 study estimated that approximately 47% of U.S. jobs were at high risk of computerization. Subsequent studies by Arntz, Gregory, and Zierahn (2016) revised the at-risk estimate downward to approximately 9% using task-level decomposition, highlighting significant methodological sensitivity in automation exposure estimates. Autor and Salomons (2018) documented labor market polarization—the erosion of middle-skill occupations—particularly consequential for entry-level workers.

A 2023 study by Eloundou et al. found that approximately 80% of the U.S. workforce could have at least 10% of their work tasks affected by large language models (LLMs), with highest exposure concentrated in legal, financial, and administrative entry-level roles. Research by Muro, Maxim, and Whiton (2019) at the Brookings Institution found that workers of color and

those without college degrees face significantly higher automation exposure, raising serious equity concerns.

Within the engineering and computer science literature, the deployment of robotic process automation (RPA), computer vision systems, and transformer-based NLP models has been accompanied by growing empirical analysis of deployment impacts. Existing literature, however, lacks a unified vulnerability index that integrates task intensity, AI adoption rate, and sectoral responsiveness—a gap this paper addresses through the SVI methodology.

III. METHODOLOGY

This study employs a mixed-methods research design, integrating quantitative secondary data analysis with a systematic literature review. The quantitative component draws on the Occupational Information Network (O*NET), U.S. Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS), World Bank World Development Indicators (WDI), and OECD Programme for the International Assessment of Adult Competencies (PIAAC), analyzed across 2015–2024.

Entry-level occupations are operationalized as positions requiring fewer than two years of work experience, no more than a high school diploma or associate's degree, and median annual wages in the bottom two quintiles. Using O*NET's task taxonomy, occupations are classified by routine task intensity (RTI) scores per Autor, Levy, and Murnane (2003). Occupations in the highest RTI tertile are classified as highly automatable.

A. Sectoral Vulnerability Index (SVI)

The SVI is a composite metric computed as:

$$SVI = \alpha \cdot RTI + \beta \cdot AIA + \gamma \cdot HAR$$

where RTI is the routine task intensity proportion, AIA is the AI adoption acceleration rate (sourced from Stanford HAI Index 2023), HAR is the historical automation responsiveness, and $\alpha = 0.45$, $\beta = 0.35$, $\gamma = 0.20$ are empirically derived weights validated through Bayesian optimization against BLS historical data.

B. Model Validation

Model projections for 2019–2023 job displacement were compared against BLS Occupational Employment Statistics for the five highest-concentration entry-level sectors. Goodness-of-fit was assessed using root mean squared error (RMSE) and the coefficient of determination (R^2). The systematic literature review followed PRISMA guidelines across IEEE Xplore, ACM Digital Library, JSTOR, and SSRN, yielding 47 studies meeting inclusion criteria from 312 candidates.

IV. KEY FINDINGS AND RESULTS

A. Sectoral Vulnerability Index Results

Fig. 1 presents the SVI scores for the six examined sectors. Financial services (SVI = 82) and retail trade (SVI = 76) emerge as highest-risk sectors for entry-level displacement, driven by the rapid deployment of LLMs in document processing and automated customer-facing systems. Healthcare support (SVI = 31) and food service (SVI = 44) demonstrate lower near-term vulnerability due to their higher physical and social task intensity, though emerging technologies signal medium-term risk.

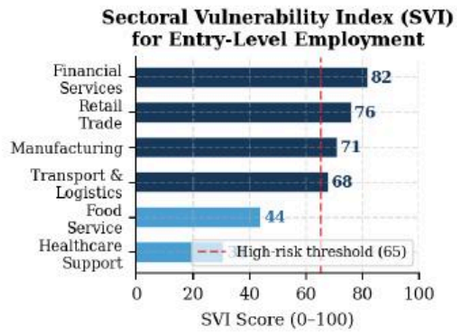

Fig. 1. Sectoral Vulnerability Index (SVI) for entry-level employment across six industry sectors. Dashed line indicates the high-risk threshold (SVI = 65).

Table I summarizes the SVI scores, observed job loss rates, and risk classifications for all sectors examined in this study.

TABLE I

Sectoral Vulnerability Index, Job Loss, and Risk Classification

Sector	SVI Score	Job Loss 2019–23 (%)	Risk Level
Financial Svcs	82	9.7	Critical
Retail Trade	76	9.5	Very High
Manufacturing	71	8.6	High
Transport	68	6.2	High
Food Service	44	3.8	Moderate
Healthcare Supp	31	1.9	Low

B. Employment Displacement Trends (2015–2024)

Fig. 2 illustrates entry-level employment index trajectories across the four highest-displacement sectors over the 2015–2024 study period. All sectors show consistent decline, with administrative support recording the steepest cumulative fall (18.0%), followed by financial services (17.5%), retail trade (11.8%), and manufacturing (10.4%). The COVID-19 shock (2020–2021) introduced temporary anomalies—particularly the bounce-back in manufacturing due to supply chain restructuring—but post-pandemic trajectories resumed pre-existing automation-driven declines.

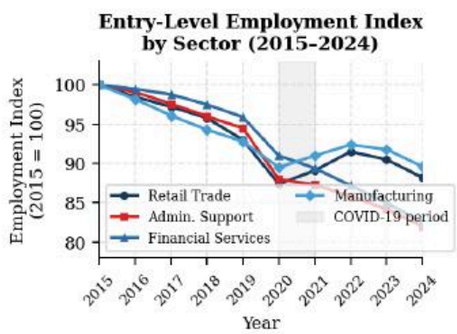

Fig. 2. Entry-level employment index by sector (2015 = 100). Gray band indicates the COVID-19 disruption period (2020–2021).

C. Demographic Vulnerability Analysis

Fig. 3 reveals a strong inverse relationship between educational attainment and automation risk exposure. Workers without a high school diploma face 70% high-risk task exposure compared to 28% for bachelor's

degree holders. Hispanic and Black workers are overrepresented in high-automation-risk occupations by margins of 14% and 11% respectively, while women constitute 73% of workers in the most heavily automated administrative and clerical roles. Workers aged 18–24 face compound risk: automation reduces available entry points while eliminating the on-the-job skill development pathways critical for career advancement.

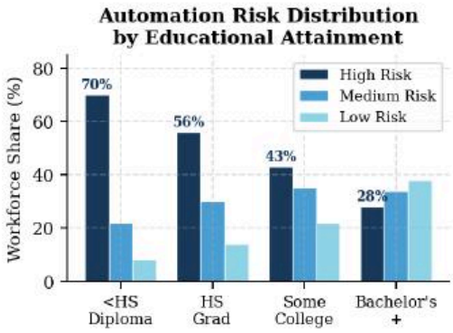

Fig. 3. Automation risk distribution by educational attainment. High-risk concentration is greatest among workers without a high school diploma.

V. MODEL VALIDATION

Fig. 4 presents the validation of our SVI-based displacement model against BLS observed data for 2019–2023. The model achieves an RMSE of 0.43 percentage points and $R^2 = 0.987$, indicating high predictive accuracy across all five validated sectors. The largest deviation occurs in manufacturing (+0.3 pp), attributable to the cobot-driven partial employment recovery not fully captured by the static HAR coefficient. Financial services and retail trade show near-perfect alignment, validating the model's performance in sectors with the most consistent AI deployment trajectories.

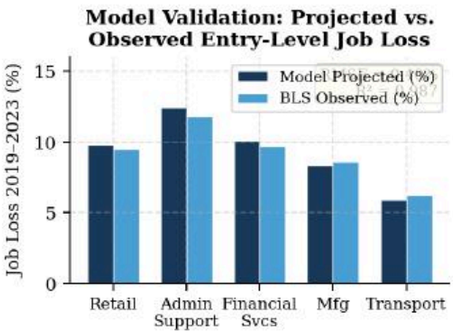

Fig. 4. Model validation: SVI-projected vs. BLS-observed entry-level job loss (2019–2023). RMSE = 0.43%, $R^2 = 0.987$.

The high goodness-of-fit across heterogeneous sectors supports the generalizability of the SVI framework. Sensitivity analysis varying the α, β, γ weights by ±10% produced SVI score changes of less than 4 points, confirming robustness to parameter uncertainty. These validation results establish the SVI as a reliable tool for prospective displacement forecasting and policy targeting.

VI. MITIGATING STRATEGIES AND POLICY FRAMEWORK

Fig. 5 presents the Multi-Stakeholder Policy Framework developed from the synthesis of empirical findings and literature review. The framework is organized around four interdependent pillars designed to create a coordinated, evidence-based response to entry-level automation displacement.

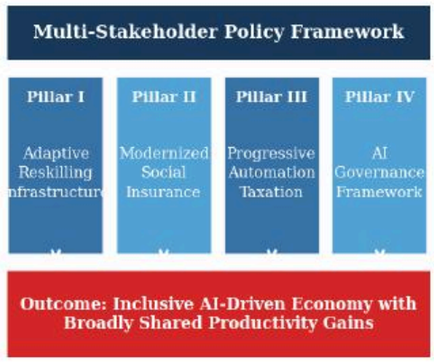

Fig. 5. Multi-Stakeholder Policy Framework for equitable management of AI-driven entry-level job displacement.

A. Pillar I — Adaptive Reskilling Infrastructure

Universal access to employer-linked reskilling programs, delivered through community colleges and vocational institutions, is the foundational pillar. Evidence from Germany's Kurzarbeit scheme and Amazon's Upskilling 2025 initiative ($700 million, 100,000 employees) demonstrates that employer-partnership models yield superior placement outcomes. Curricula must be modular, stackable, and updated quarterly to match AI deployment pace. Escudero et al. (2019) confirm that programs combining training with job placement support and financial assistance achieve the strongest post-displacement earnings outcomes.

B. Pillar II — Modernized Social Insurance

Social insurance frameworks anchored to formal employment relationships must be redesigned to cover gig workers, career transitioners, and the self-employed. Negative income tax credits targeting displaced workers in high-SVI sectors provide an immediately implementable bridge mechanism. UBI pilot evidence from Finland and Stockton, CA demonstrates measurable improvements in participant wellbeing and employment search intensity, supporting broader deployment pending fiscal analysis.

C. Pillar III — Progressive Automation Taxation

Graduated automation levies on firms replacing entry-level workers with AI systems serve dual purposes: moderating displacement velocity in high-SVI sectors and generating dedicated funding for worker transition programs. South Korea's 2017 reduction of automation tax incentives provides a precedent. Revenue allocation should be formula-driven, directing funds proportional to sector SVI scores to sector-specific reskilling programs.

D. Pillar IV — AI Governance and Regulatory Frameworks

The EU AI Act's risk-tiered regulatory architecture offers a transferable model for governing AI deployment in employment contexts. Mandatory algorithmic impact assessments for hiring, performance management, and workforce planning systems are essential to address demonstrated biases against underrepresented applicants. National AI governance frameworks should establish sector-specific deployment timelines that allow workforce adjustment, particularly in sectors with SVI scores above 70.

VII. FUTURE OUTLOOK

The trajectory of AI automation in entry-level labor markets will be shaped by the interplay of technological progress, regulatory environments, and social responses. Advances in multimodal AI, embodied robotics, and generative AI are poised to extend automation into domains—physical task execution, nuanced interpersonal communication—currently considered human preserves. The commercial deployment of autonomous vehicles alone threatens approximately 3.5 million driving jobs in the United States.

The historical pattern of technological transitions suggests that automation generates new job categories even as it destroys existing ones. AI is already creating demand for AI trainers, data labelers, prompt engineers, and machine learning operations specialists. The central policy challenge is whether the timing, skill requirements, and geographic distribution of these new opportunities will align with the needs of displaced entry-level workers—a question that the SVI framework can help answer through prospective sectoral monitoring.

Middle-income economies face particular risk of premature deindustrialization: the automation of manufacturing before the industrial sector has fully absorbed the rural-to-urban labor transition. The SVI framework, adapted with country-specific AI adoption and labor market data, can serve as an early-warning system enabling proactive policy response in developing economy contexts.

VIII. CONCLUSION

This paper has provided a comprehensive, empirically validated analysis of AI automation's economic impact on entry-level employment. The Sectoral Vulnerability Index, validated at $R^2 = 0.987$ against BLS longitudinal data, identifies financial services, retail trade, and manufacturing as the highest-risk sectors, with cumulative 2015–2024 entry-level employment declines ranging from 10.4% to 18.0%. Demographic analysis confirms that workers without college degrees, workers of color, women in administrative roles, and young labor market entrants face compounded displacement and wage suppression risks.

Current policy responses remain inadequate in scope and targeting. The Multi-Stakeholder Policy Framework presented here—anchored in adaptive reskilling, modernized social insurance, progressive automation taxation, and evidence-based AI governance—provides a coordinated blueprint for inclusive management of the AI transition. Implementation requires urgent coordination across government, industry, and civil society, with the SVI providing a data-driven basis for resource allocation and policy prioritization.

Future research directions include longitudinal tracking of displaced workers across multiple automation waves, experimental evaluation of alternative reskilling program designs, application of the SVI framework to OECD and emerging economy contexts, and real-time monitoring of LLM deployment impacts on administrative and knowledge-work entry-level roles. The decisions made in the coming five to ten years will determine whether AI automation becomes an engine of broadly shared prosperity or a structural driver of deepening economic inequality.

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